Architecture 1001: x86-64 Assembly TooManyParameters.c with GCC

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TooManyParameters.c

```c
#define uint64 unsigned long long

int func(uint64 a, uint64 b, uint64 c, uint64 d, uint64 e){
    int i = a+b-c+d-e;
    return i;
}
int main(){
    return func(0x11,0x22,0x33,0x44,0x55);
}
```

```
main:
 endbr64
 push   %rbp
 mov    %rsp,%rbp
 mov    $0x55,%r8d
 mov    $0x44,%ecx
 mov    $0x33,%edx
 mov    $0x22,%esi
 mov    $0x11,%edi
 callq  0x555555555129 <func>
 pop    %rbp
 retq
```

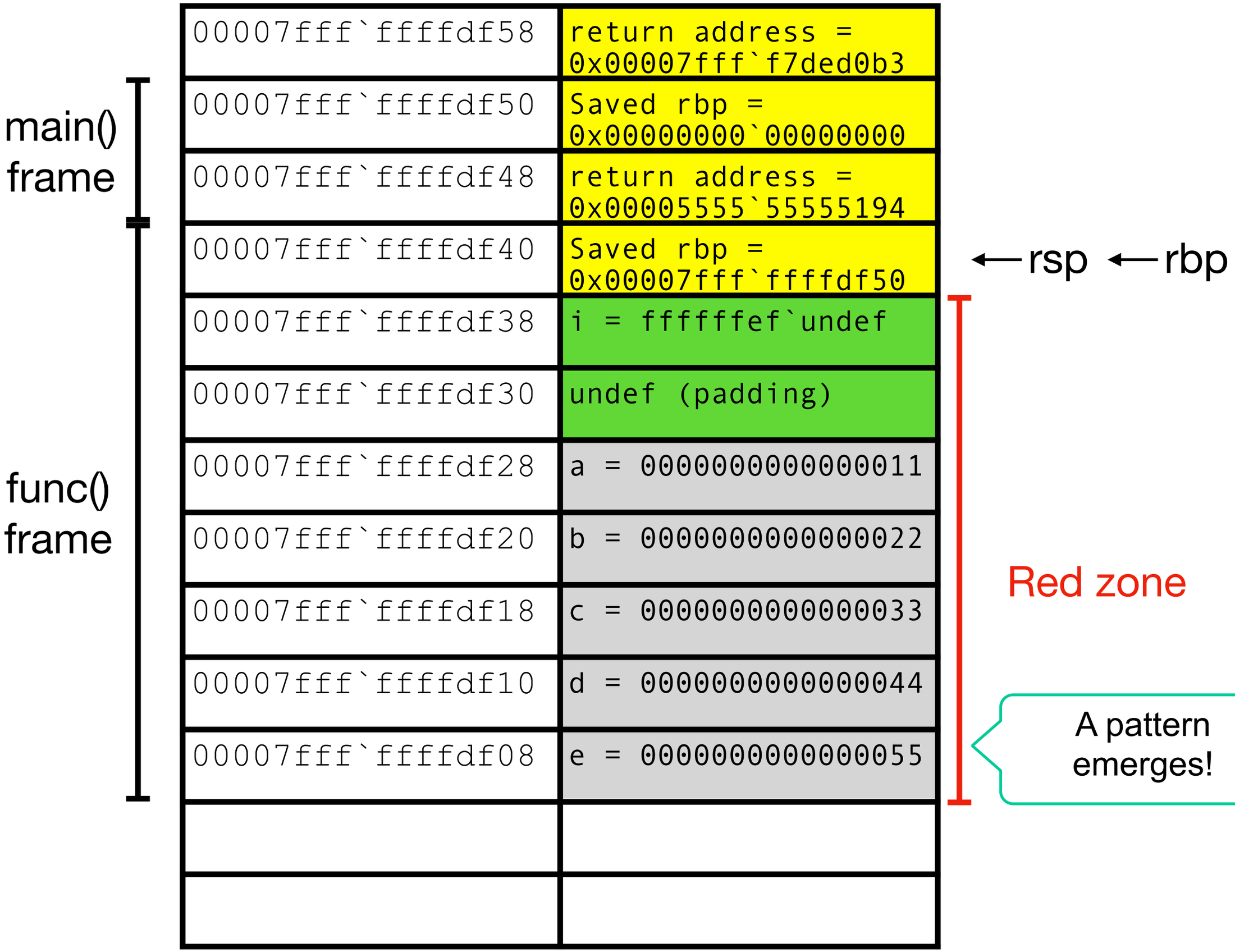
```
func:
0x0000555555555129 <+0>:  endbr64
0x000055555555512d <+4>:  push   %rbp
0x000055555555512e <+5>:  mov    %rsp,%rbp
0x0000555555555131 <+8>:  mov    %rdi,-0x18(%rbp)
0x0000555555555135 <+12>: mov    %rsi,-0x20(%rbp)
0x0000555555555139 <+16>: mov    %rdx,-0x28(%rbp)
0x000055555555513d <+20>: mov    %rcx,-0x30(%rbp)
0x0000555555555141 <+24>: mov    %r8,-0x38(%rbp)
0x0000555555555145 <+28>: mov    -0x18(%rbp),%rax
0x0000555555555149 <+32>: mov    %eax,%edx
0x000055555555514b <+34>: mov    -0x20(%rbp),%rax
0x000055555555514f <+38>: add    %eax,%edx
0x0000555555555151 <+40>: mov    -0x28(%rbp),%rax
0x0000555555555155 <+44>: sub    %eax,%edx
0x0000555555555157 <+46>: mov    -0x30(%rbp),%rax
0x000055555555515b <+50>: add    %eax,%edx
0x000055555555515d <+52>: mov    -0x38(%rbp),%rax
0x0000555555555161 <+56>: sub    %eax,%edx
0x0000555555555163 <+58>: mov    %edx,%eax
0x0000555555555165 <+60>: mov    %eax,-0x4(%rbp)
0x0000555555555168 <+63>: mov    -0x4(%rbp),%eax
0x000055555555516b <+66>: pop    %rbp
0x000055555555516c <+67>: retq
```

Parameter Conventions: System V “x86-64” ABI (GCC et al.)

- First ***6*** parameters (from left to right) are put into RDI, RSI, RDX, RCX, R8, R9 respectively
- Remaining parameters pushed onto the stack so that the left-most parameter is at the lowest address

```
              RDI     RSI     RDX     RCX      R8
int func(int a, int b, int c, int d, int e){
  int i = a+b-c+d-e;
  return i;
}
int main(){
  return func(0x11,0x22,0x33,0x44,0x55);
}
```

The stack looks like this at line 0x5555`55555168 in func():



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TooManyParameters.c takeaways

- GCC also stores register-passed function parameters on the stack, for debugging purposes